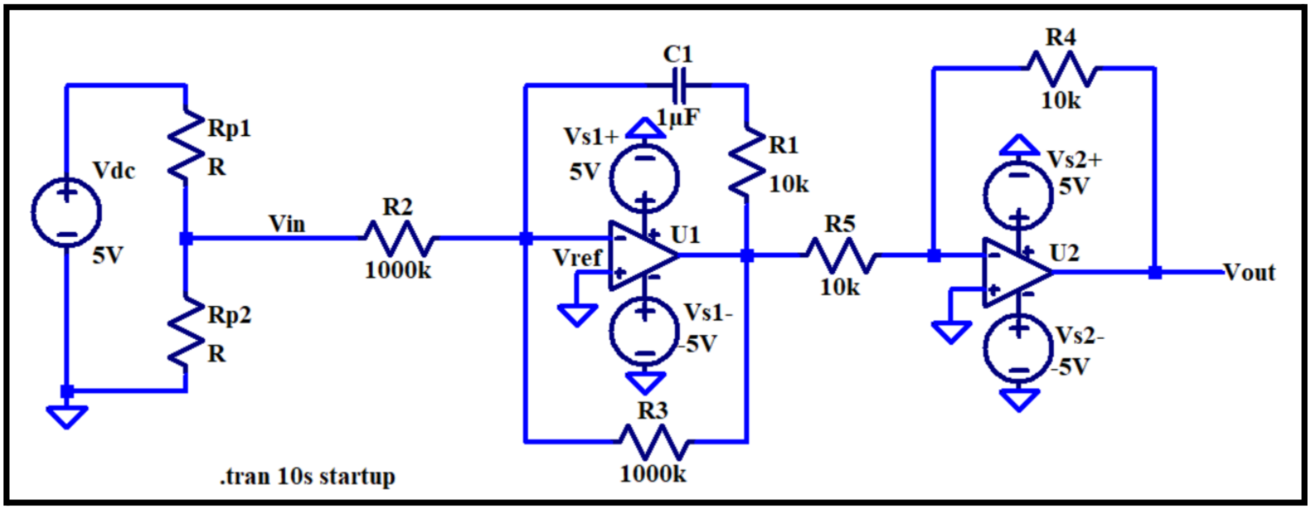
PI and PD Controllers

Building Block:



Description:

PI Controller designed with 2 inverting op amps. Vin is the input voltage controlled by a 5kΩ potentiometer represented as the voltage divider on the left with a 5V dc source given from an arduino (had to improvise since Vs for op amps occupied V+/- on the instrumentation board) and 2 resistors (Rp1 and Rp2). The voltage sources are +5V and -5V for each op amp. This is a PI controller with the incorporation of the RC in parallel with R3. This design results in Vout (voltage output) being about equal to Vin (input voltage), within around 5% error. The integral controller is C1 and proportional controller is R1.

Analysis:

Voltage divider to model voltage input, non-inverting error op amp

Vin= Vdc(Rp2/(Rp2+Rp1))

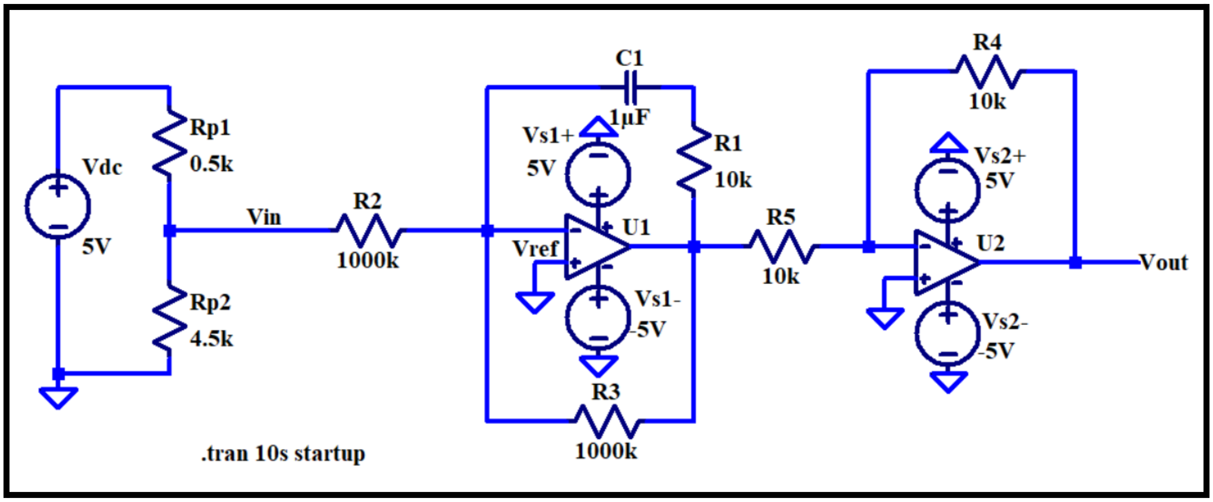
Op Amp 1- Is an integrating op amp with 3 resistors and a capacitor. The capacitor charges up until it reaches steady state where it would appear as an open circuit. This means no current will be flowing through that section of the op amp. This in essence will turn the op amp into an inverting op amp with the resistor R3 as 1000k as time goes on. This means the output voltage of this op amp will be, Vout=-Vin(R3/R2)=-Vin(1000k/1000k)=-Vin. Using this output voltage, the transfer function is calculated as H=Vout/Vin= -Vin/Vin= -1.

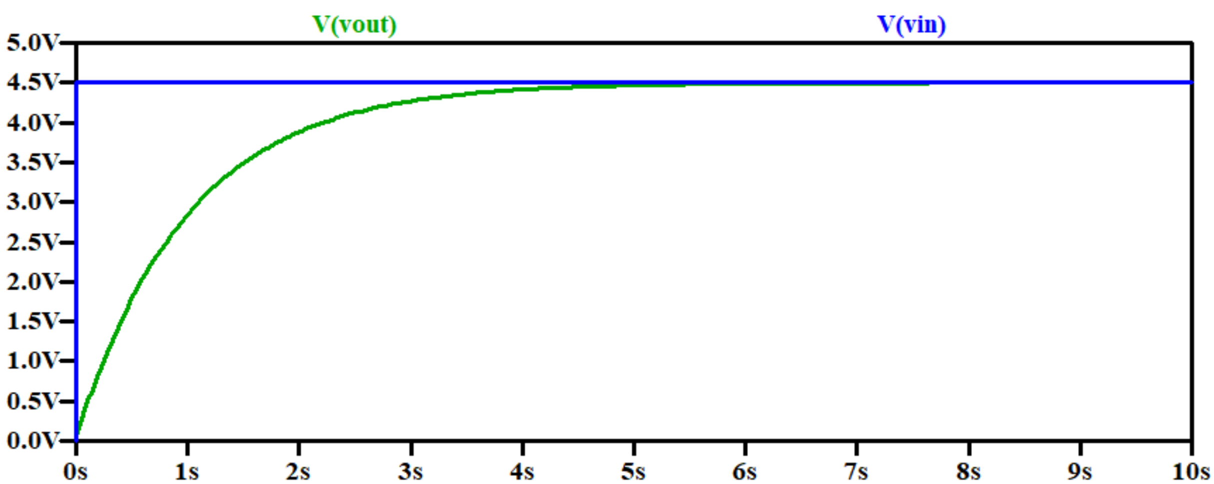
Op Amp 2- This amplifier is an inverting amplifier with two 10k resistances. The output voltage for an inverter is Vout=Vin(-Rf/Ri). In this inverter, Rf=Ri so the output voltage will just be equal to the negative input voltage. This comes out to a transfer function of -1 as Vout/Vin=-1.

The entire transfer function for the schematic will be equal to 1 as H(total)=multiplication of all transfer functions for the entire schematic. The 2 transfer function for the whole schematic were both equal to -1 meaning H(total)= (-1)(-1)=1 so the input voltage should equal the output voltage of the whole schematic; the input voltage is labeled as Vin and is calculated using a voltage divider as shown before.

Simulation:

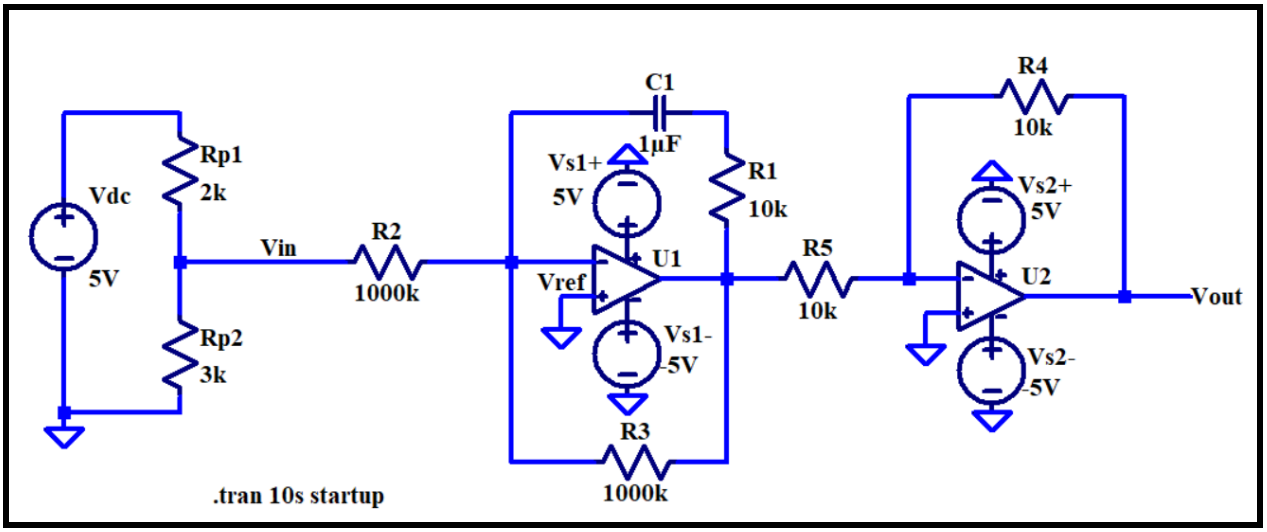
When Vin is 4.5V:

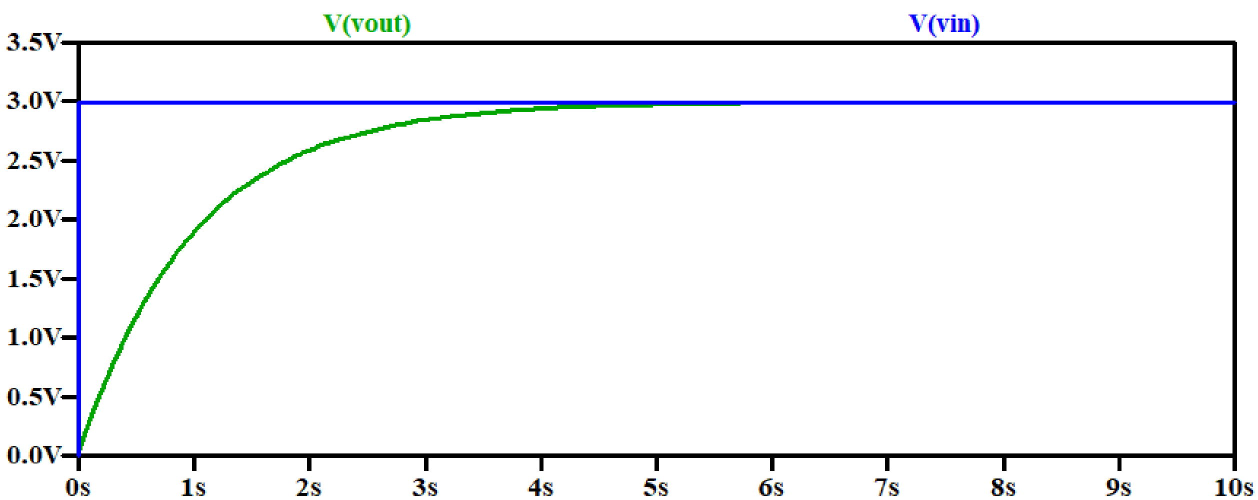




Vout is simulated to reach 4.5V

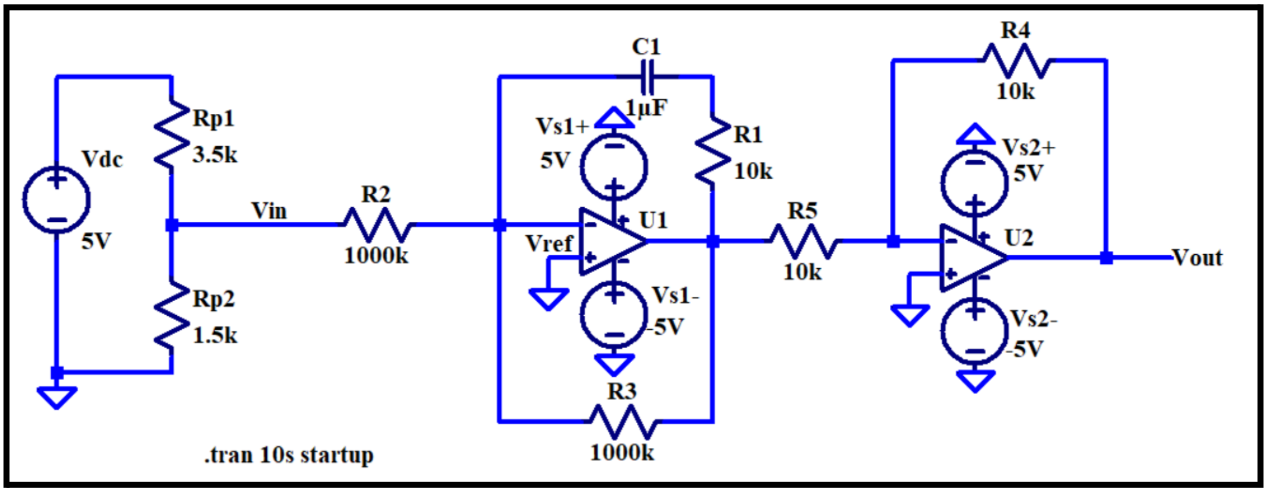
When Vin is 3V:

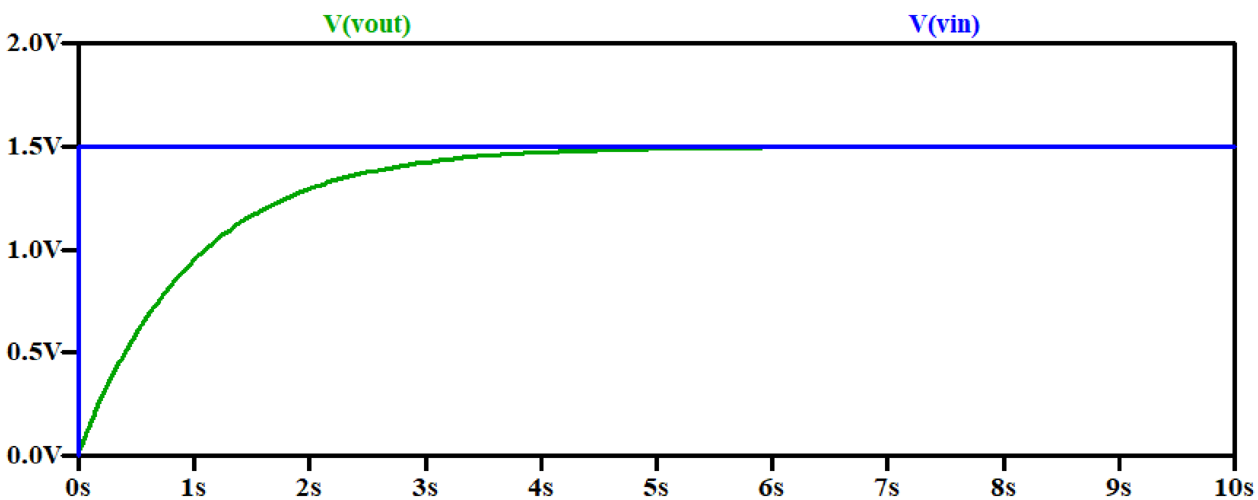




Vout is simulated to reach 3V

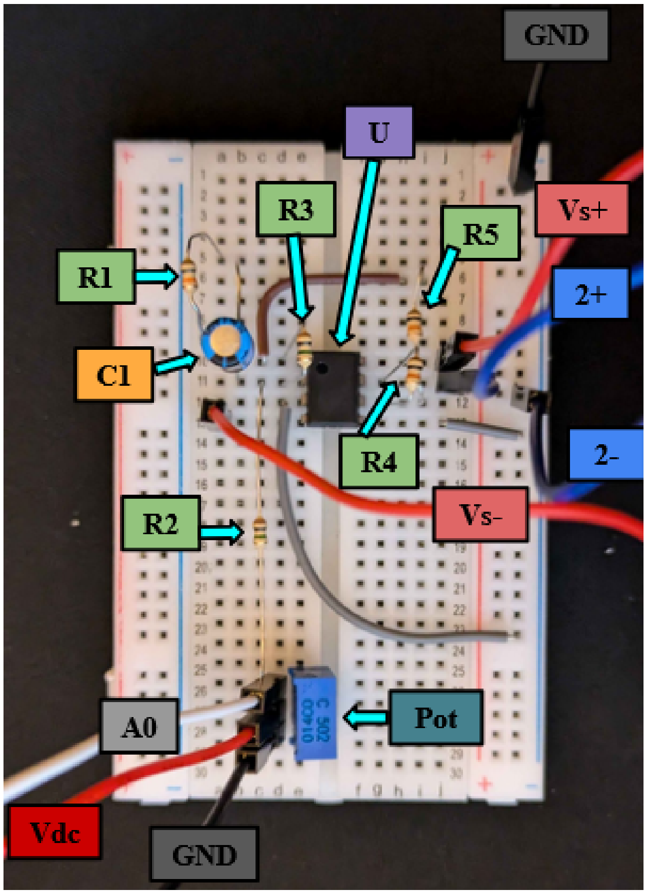
When Vin is 1.5V:





Vout is simulated to reach 1.5V

Measurement:



R1 = R4 = R5 = 10kΩ

R2 = R3 = 1MΩ

C1 = 1μF

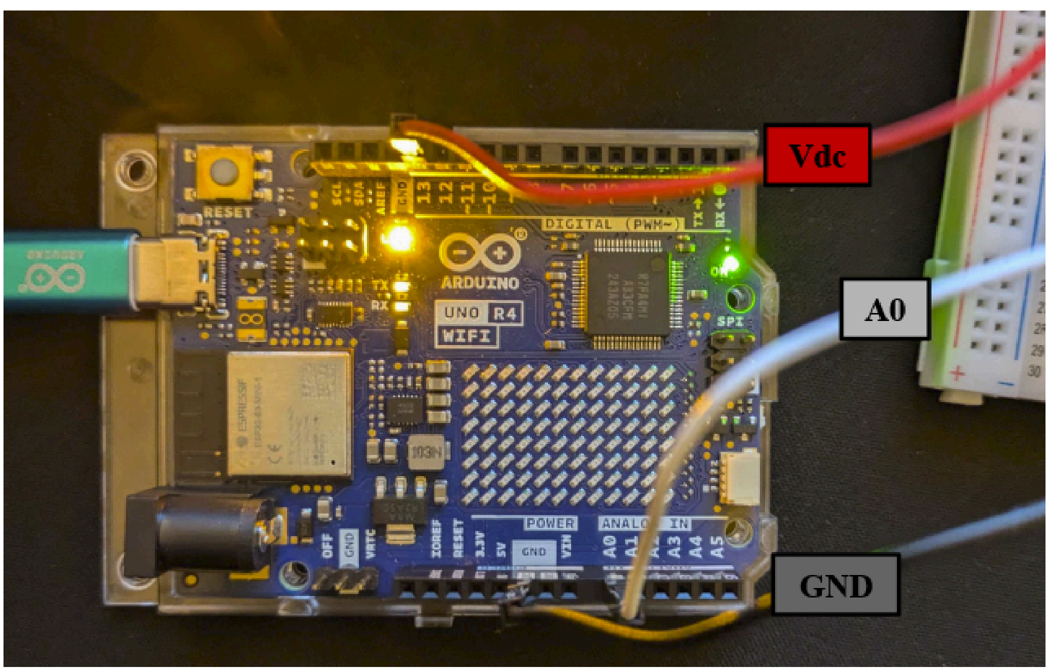
Vs+/- = +5/-5V

Vdc = +5V

U has 2 op-amps (TL072IP chip)

Pot is the 5kΩ potentiometer

Channel 2 measures Vout



On MATLAB, to create a 5V dc source and have A0 read Vin from the potentiometer:

lab2.m

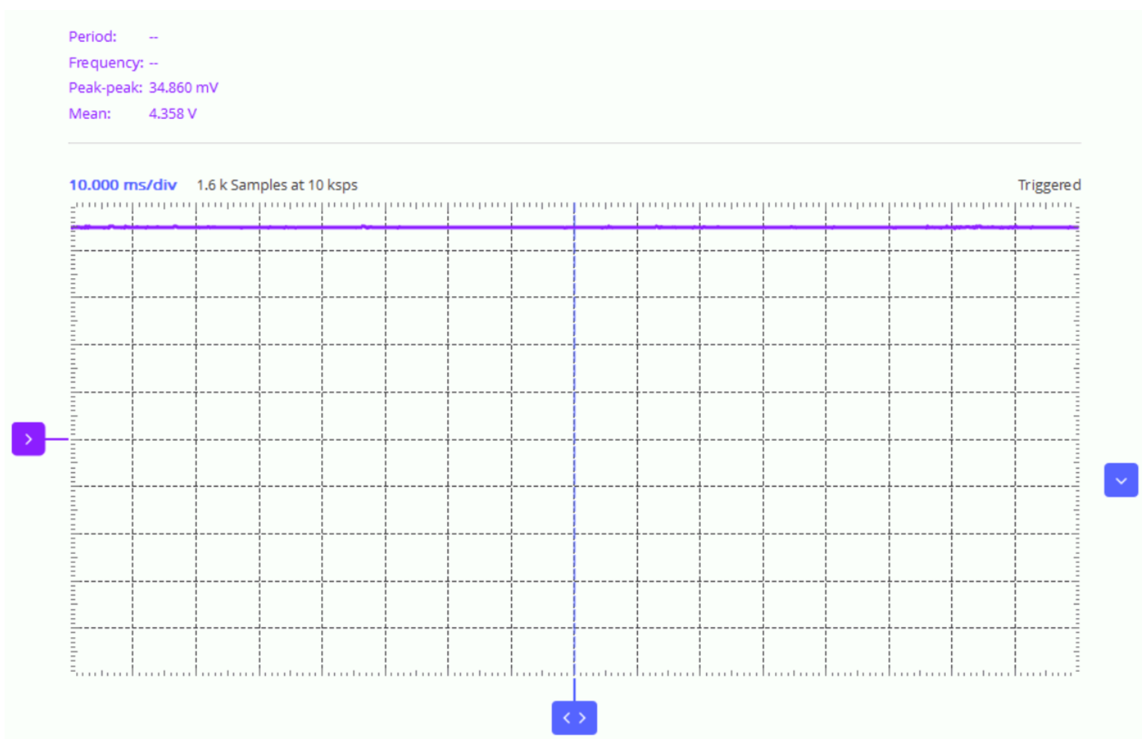
```
a = arduino();
writeDigitalPin(a,'D13',1);
t=1;
while t>0
    Vin = readVoltage(a, 'A0');
    display(Vin);
end

```

When Vin is around 4.5V on MATLAB:

```
Vin =

    4.4477
```

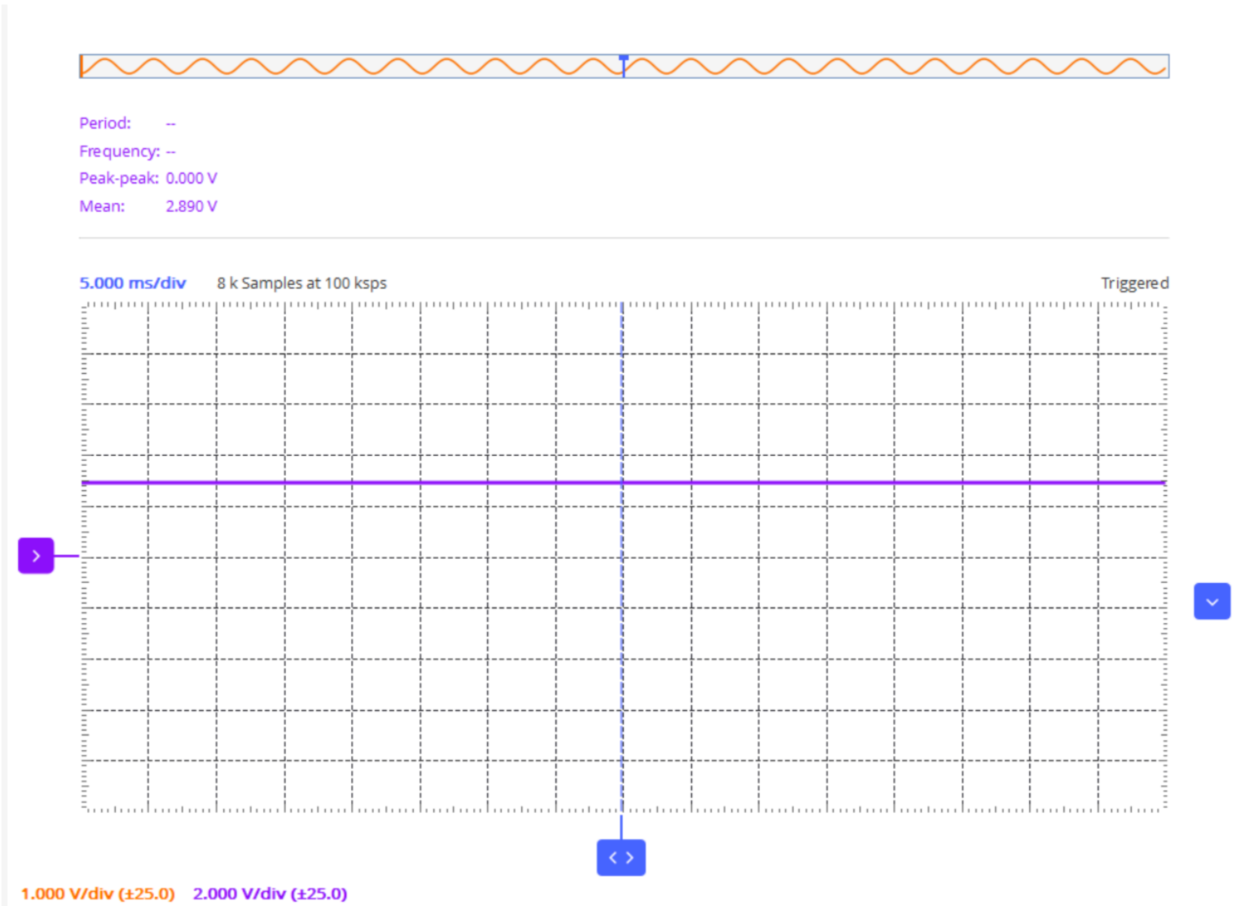


Vout is measured to be around 4.358V on scopy

When Vin is around 3V on MATLAB:

```
Vin =

    3.0547
```

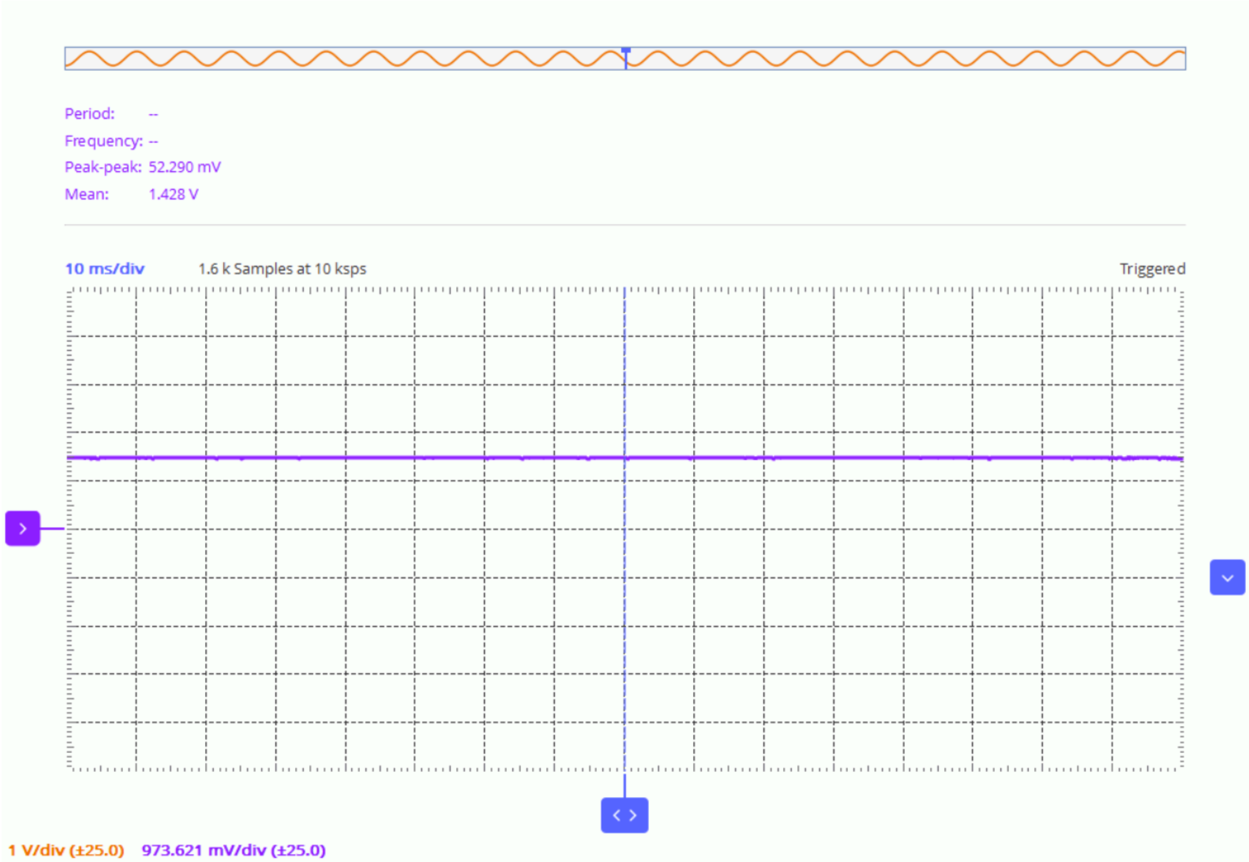


Vout is measured to be around 2.859 on scopy

When Vin is around 1.5V on MATLAB:

```
Vin =

    1.5054
```



Vout is measured to be around 1.428V on scopy.

Discussion (and answer related questions in Alpha Lab):

Vref	Analysis	Simulation	Measurement	% Err
4.5V	4.5V	4.5V	4.345V	3.44%
3V	3V	3V	2.890V	4.70%
1.5V	1.5V	1.5V	1.428V	4.80%

The percent error for each is less than 5% which is reasonable since real life measurements don't come out to be as ideal as calculations or simulations due to circuit elements not being perfectly the value they are assumed to be. However, our error is low enough to conclude that the results of measurement are the values we expected to get and therefore allow us to prove that our PI controller design successfully produces a Vout that is about the value of Vin from an adjustable voltage source (via potentiometer).